

Course Code: BCO601**Course: Microcontroller and Embedded Systems****Credits: 4****L:T:P:S 3:0:2****CIE: 25****SEE: 100****SEE Hours: 3****Max. Marks: 100****Course Outcomes:**

On successful completion of the course, the student will be able to:

COs	Course Outcomes	Bloom's level
CO1	Understand the architectural features and instruction set of 32 bit ARM microcontrollers.	L2
CO2	Apply instructions of assembly language for programming ARM.	L3
CO3	Interpret the basic hardware components and their selection method based on the characteristics and attributes of an embedded system.	L3
CO4	Explain the need of real time operating system for embedded system applications.	L2
CO5	Develop/test/Conduct the experiments on an ARM7TDMI/LPC2148 evaluation board using Embedded 'C' and Keil Vision tool/Compiler	L3

Mapping with POs and PSOs:**Mapping with POs and PSOs:**

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12		PSO1	PSO2
CO1	3	-	-	-	-	-	-	1	-	-	-	-		2	2
CO2	-	3	3	3	3	3	2	-	1	-	3	-		3	3
CO3	3	2	-	-	2	-	-	-	-	-	-	-		2	2
CO4	3	3	3	3	3	2	2	-	-	-	-	-		2	2
CO5	3										3			3	3

Mapping Strength: Strong– 3 Medium – 2 Low – 1

Course Structure

	Module – 1 -	No. of Lecture Hours	No. of Tutorial Hours	Self Learning Hours
1.1	Microprocessors versus Microcontrollers, ARM Embedded Systems: The RISC design philosophy, The ARM Design Philosophy,	2	-	-
1.2	Embedded System Hardware, Embedded System Software,	2	-	-
1.3	ARM Processor Fundamentals: Registers, Current Program Status Register,	2	-	-
1.4	Pipeline, Exceptions, Interrupts, and the Vector Table , Core Extensions.	2	-	-
	Module – 2 -		-	-
2.1	Introduction to the ARM Instruction Set : Data Processing Instructions ,	2	-	-
2.2	Programme Instructions, Software Interrupt Instructions,	2	-	-
2.3	Program Status Register Instructions, Coprocessor Instructions, Loading Constants	2	-	-
2.4	ARM programming using Assembly language: Writing Assembly code, Profiling and cycle counting, instruction scheduling,	2		
2.5	Register Allocation, Conditional Execution, Looping Constructs	2		
	Module – 3 -		-	-
3.1	Embedded Vs General computing system, History of embedded systems, Classification of Embedded systems,	2	-	-
3.2	Major applications areas of embedded systems, purpose of embedded systems Core of an Embedded System including all types of processor/controller,	2	-	-
3.3	Memory, Sensors, Actuators, LED, 7 segment LED display,	2	-	-
3.4	Stepper motor, Keyboard , Push button switch.	2		
	Module – 4 -		-	-
4.1	Characteristics and Quality Attributes of Embedded Systems,	2	-	-
4.2	Operational quality attributes, nonoperational quality attributes,	2	-	-
4.3	Embedded Systems-Application and Domain specific	2	-	-
4.4	Hardware Software Co-Design and Program Modelling.	1	-	-
	Module – 5 -		-	-
5.1	Operating System basics, Types of operating systems,	2	-	-

5.2	Task, process and threads (Only POSIX Threads with an example program), Thread preemption, Multiprocessing and Multitasking, Task Communication (without any program),	2	-	-
5.3	Task synchronization issues – Racing and Deadlock, Concept of Binary and counting semaphores (Mutex example without any program),	2	-	-
5.4	How to choose an RTOS , Integration and testing of Embedded hardware and firmware.	1	-	-
<i>Total No. of Lecture Hours</i>		40		
<i>Total No. of Tutorial Hours</i>			0	
<i>Total No. of Self learning Hours</i>				0

Text Books:

1. Andrew N Sloss, Dominic Symes and Chris Wright, ARM system developers guide, Elsevier, Morgan Kaufman publishers, 2008.
2. Shibu K V, “Introduction to Embedded Systems”, Tata McGraw Hill Education, Private Limited, 2 nd Edition.

Reference Books:

1. Raghunandan..G.H, Microcontroller (ARM) and Embedded System, Cengage learning Publication, 2019
2. The Insider’s Guide to the ARM7 Based Microcontrollers, Hitex Ltd., 1st edition, 2005.
3. Steve Furber, ARM System-on-Chip Architecture, Second Edition, Pearson, 2015.
4. Raj Kamal, Embedded System, Tata McGraw-Hill Publishers, 2nd Edition, 2008.

Online Resources:

1. <https://www.coursera.org/specializations/cortex-m-architecture-and-software-development>
2. <https://www.coursera.org/programs/faculty-learning-program-iqr5x/learn/introduction-embedded-systems?source=search>

Lab Experiments – 15 Hours

Conduct the following experiments by writing programs using ARM7TDMI/LPC2148 using an evaluation board/simulator/evaluation version of Embedded 'C' & Keil Uvision-4 tool/compiler. and the required software tool.

1. Develop a program to multiply two 16 bit binary numbers.
2. Write a program to find the sum of first 10 integer numbers.
3. Write a program to find factorial of a number.
4. Write a program to add an array of 16 bit numbers and store the 32 bit result in internal RAM
5. Write a program to find the square of a number (1 to 10) using look-up table.
6. Write a program to find the largest/smallest number in an array of 32 numbers .
7. Display “Hello World” message using Internal UART.
8. Interface a Stepper motor and rotate it in clockwise and anti-clockwise direction
9. Display the Hex digits 0 to F on a 7-segment LED interface, with an appropriate delay in between
10. Interface a 4x4 keyboard and display the key code on an LCD.